Ethereum & Hyperledger Fabric (Part-02)

Class Test Practice Set - Smart Contract Deployment, Update and ETH Transfer Fees (10 Marks each)

Formula Sheet (from lecture slides)

Transaction Fee	Gas Used × Gas Price
Effective Gas Price	Base Fee + Priority Fee (Tip)
Total Fee	Gas Used × (Base Fee + Priority Fee)
Burned Amount	Gas Used × Base Fee
Validator Reward	Gas Used × Priority Fee
Units	1 Gwei = 10^9 Wei = 0.000000001 ETH; 1 ETH = 10^9 Gwei
Standard ETH transfer	21,000 gas units (Gas Limit unused is refunded)
Sender pays (transfer)	Value sent + Transaction Fee

Pattern of every question: (1) Deploy a Smart Contract, find **Tx1** cost; (2) Update the contract, find **Tx2** cost; (3) **Tx3**: A sends 2 ETH to B. Show all steps and give final answers in ETH. Each question carries 10 marks.

Question 1 [10 Marks]

Ahsan (Account A) works on Ethereum using MetaMask.

(a) Ahsan deploys a smart contract (**Tx1**). He sets a Gas Limit of 100,000 units. The contract deployment uses 85,000 gas. The network Base Fee is 30 Gwei and he sets a Priority Fee of 3 Gwei. Calculate (i) the Tx1 cost in ETH, (ii) the ETH burned, and (iii) the validator reward. **[4]**

(b) He then updates the contract state (**Tx2**). Gas Limit = 50,000, Gas Used = 42,000, Base Fee = 25 Gwei, Priority Fee = 3 Gwei. Calculate the Tx2 fee in ETH. **[3]**

(c) Finally, A sends **2 ETH** to B (**Tx3**, a standard transfer). Base Fee = 20 Gwei, Priority Fee = 2 Gwei. Find the Tx3 fee, the total ETH deducted from A, and the ETH received by B. **[3]**

Answer

(a) [4 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 30 + 3 = **33 Gwei**
- **Tx1 Fee** = Gas Used × Effective Gas Price = 85,000 × 33 = 2,805,000 Gwei = **0.002805 ETH**
- **Burned** = Gas Used × Base Fee = 85,000 × 30 = 2,550,000 Gwei = **0.00255 ETH**
- **Validator Reward** = Gas Used × Priority Fee = 85,000 × 3 = 255,000 Gwei = **0.000255 ETH**

(b) [3 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 25 + 3 = **28 Gwei**
- **Tx2 Fee** = Gas Used × Effective Gas Price = 42,000 × 28 = 1,176,000 Gwei = **0.001176 ETH**
- **Burned** = Gas Used × Base Fee = 42,000 × 25 = 1,050,000 Gwei = **0.00105 ETH**
- **Validator Reward** = Gas Used × Priority Fee = 42,000 × 3 = 126,000 Gwei = **0.000126 ETH**

(c) [3 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 20 + 2 = **22 Gwei**
- **Tx3 Fee** = Gas Used × Effective Gas Price = 21,000 × 22 = 462,000 Gwei = **0.000462 ETH**
- **Burned** = Gas Used × Base Fee = 21,000 × 20 = 420,000 Gwei = **0.00042 ETH**
- **Validator Reward** = Gas Used × Priority Fee = 21,000 × 2 = 42,000 Gwei = **0.000042 ETH**
- **Total deducted from A** = Value + Fee = 2 + 0.000462 = **2.000462 ETH**
- **Received by B** = **2 ETH** (the fee is paid by the sender, not by the receiver)

Question 2 [10 Marks]

Nadia (Account A) interacts with a voting dApp on Ethereum.

(a) Nadia deploys a voting smart contract (**Tx1**). Gas Limit = 150,000 and Gas Used = 120,000. The Base Fee is 50 Gwei and the Priority Fee is 5 Gwei. Calculate (i) the total Tx1 cost in ETH, (ii) the burned amount, and (iii) the amount paid to the validator. **[4]**

(b) Later she updates the contract (**Tx2**) to add a new candidate. Gas Limit = 75,000, Gas Used = 60,000, Base Fee = 40 Gwei, Priority Fee = 4 Gwei. Calculate the Tx2 cost in ETH and state how much of it is burned. **[3]**

(c) She also sends **2 ETH** from A to B (**Tx3**). Base Fee = 35 Gwei, Priority Fee = 3 Gwei. Calculate the total ETH deducted from A and the validator reward for this transfer. **[3]**

Answer

(a) [4 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 50 + 5 = **55 Gwei**
- **Tx1 Fee** = Gas Used × Effective Gas Price = 120,000 × 55 = 6,600,000 Gwei = **0.0066 ETH**
- **Burned** = Gas Used × Base Fee = 120,000 × 50 = 6,000,000 Gwei = **0.006 ETH**
- **Validator Reward** = Gas Used × Priority Fee = 120,000 × 5 = 600,000 Gwei = **0.0006 ETH**

(b) [3 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 40 + 4 = **44 Gwei**
- **Tx2 Fee** = Gas Used × Effective Gas Price = 60,000 × 44 = 2,640,000 Gwei = **0.00264 ETH**
- **Burned** = Gas Used × Base Fee = 60,000 × 40 = 2,400,000 Gwei = **0.0024 ETH**

(c) [3 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 35 + 3 = **38 Gwei**
- **Tx3 Fee** = Gas Used × Effective Gas Price = 21,000 × 38 = 798,000 Gwei = **0.000798 ETH**
- **Validator Reward** = Gas Used × Priority Fee = 21,000 × 3 = 63,000 Gwei = **0.000063 ETH**
- **Total deducted from A** = Value + Fee = 2 + 0.000798 = **2.000798 ETH**

Question 3 [10 Marks]

Rafi (Account A) starts with a balance of **5 ETH**. B is Sumi's account.

(a) Rafi deploys a token contract (**Tx1**): Gas Limit = 200,000, Gas Used = 175,000, Base Fee = 18 Gwei, Priority Fee = 2 Gwei. Calculate the Tx1 cost in ETH, the burned amount, and the validator reward. **[4]**

(b) Rafi then updates the contract (**Tx2**): Gas Limit = 90,000, Gas Used = 72,000, Base Fee = 22 Gwei, Priority Fee = 3 Gwei. Calculate the Tx2 cost in ETH. **[3]**

(c) Rafi finally transfers **2 ETH** to B (**Tx3**): Base Fee = 25 Gwei, Priority Fee = 1 Gwei. Calculate the fee of Tx3 and find A's final balance after all three transactions. **[3]**

Answer

(a) [4 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 18 + 2 = **20 Gwei**
- **Tx1 Fee** = Gas Used × Effective Gas Price = 175,000 × 20 = 3,500,000 Gwei = **0.0035 ETH**
- **Burned** = Gas Used × Base Fee = 175,000 × 18 = 3,150,000 Gwei = **0.00315 ETH**
- **Validator Reward** = Gas Used × Priority Fee = 175,000 × 2 = 350,000 Gwei = **0.00035 ETH**

(b) [3 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 22 + 3 = **25 Gwei**
- **Tx2 Fee** = Gas Used × Effective Gas Price = 72,000 × 25 = 1,800,000 Gwei = **0.0018 ETH**
- **Burned** = Gas Used × Base Fee = 72,000 × 22 = 1,584,000 Gwei = **0.001584 ETH**
- **Validator Reward** = Gas Used × Priority Fee = 72,000 × 3 = 216,000 Gwei = **0.000216 ETH**

(c) [3 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 25 + 1 = **26 Gwei**
- **Tx3 Fee** = Gas Used × Effective Gas Price = 21,000 × 26 = 546,000 Gwei = **0.000546 ETH**
- **Total fees** = 3,500,000 + 1,800,000 + 546,000 = 5,846,000 Gwei = 0.005846 ETH
- **Final balance of A** = 5 – 2 (sent to B) – 0.005846 (fees) = **2.994154 ETH**

Question 4 [10 Marks]

Mitu (Account A) is working with a certificate-issuing smart contract. Remember that unused gas is refunded.

(a) Mitu deploys the contract (**Tx1**). Gas Limit = 120,000, Gas Used = 96,000, Base Fee = 45 Gwei, Priority Fee = 5 Gwei. Calculate (i) the maximum fee MetaMask reserves (Gas Limit × Effective Gas Price), (ii) the actual Tx1 fee, (iii) the refund for unused gas, and (iv) the amount burned. **[4]**

(b) She updates the contract (**Tx2**). Gas Limit = 60,000, Gas Used = 38,000, Base Fee = 35 Gwei, Priority Fee = 2 Gwei. Calculate the Tx2 fee and the validator reward. **[3]**

(c) Mitu sends **2 ETH** from A to B (**Tx3**). Base Fee = 28 Gwei, Priority Fee = 2 Gwei. Calculate the Tx3 fee and the total ETH deducted from A. **[3]**

Answer

(a) [4 marks]

- **Effective Gas Price** = 45 + 5 = **50 Gwei**
- (i) **Max fee reserved** = 120,000 × 50 = 6,000,000 Gwei = **0.006 ETH**
- (ii) **Actual fee** = 96,000 × 50 = 4,800,000 Gwei = **0.0048 ETH**
- (iii) **Refund** = unused gas × effective price = (120,000 – 96,000) × 50 = 24,000 × 50 = 1,200,000 Gwei = **0.0012 ETH** (check: 0.006 – 0.0048 = 0.0012)
- (iv) **Burned** = 96,000 × 45 = 4,320,000 Gwei = **0.00432 ETH**

(b) [3 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 35 + 2 = **37 Gwei**
- **Tx2 Fee** = Gas Used × Effective Gas Price = 38,000 × 37 = 1,406,000 Gwei = **0.001406 ETH**
- **Validator Reward** = Gas Used × Priority Fee = 38,000 × 2 = 76,000 Gwei = **0.000076 ETH**

(c) [3 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 28 + 2 = **30 Gwei**
- **Tx3 Fee** = Gas Used × Effective Gas Price = 21,000 × 30 = 630,000 Gwei = **0.00063 ETH**
- **Total deducted from A** = Value + Fee = 2 + 0.00063 = **2.00063 ETH**

Question 5 [10 Marks]

Tanvir (Account A) performs three consecutive transactions on Ethereum and wants to analyse the total cost.

(a) Tx1 (deploy a smart contract): Gas Used = 110,000, Base Fee = 32 Gwei, Priority Fee = 4 Gwei. Calculate the Tx1 cost in ETH. **[3]**

(b) Tx2 (update the contract): Gas Used = 55,000, Base Fee = 36 Gwei, Priority Fee = 4 Gwei. Calculate the Tx2 cost in ETH. **[3]**

(c) Tx3: A sends **2 ETH** to B. Base Fee = 30 Gwei, Priority Fee = 3 Gwei. Calculate the Tx3 fee and the total ETH leaving A's account. **[2]**

(d) Across all three transactions, find the total ETH burned and the total ETH paid to validators. **[2]**

Answer

(a) [3 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 32 + 4 = **36 Gwei**
- **Tx1 Fee** = Gas Used × Effective Gas Price = 110,000 × 36 = 3,960,000 Gwei = **0.00396 ETH**
- **Burned** = Gas Used × Base Fee = 110,000 × 32 = 3,520,000 Gwei = **0.00352 ETH**
- **Validator Reward** = Gas Used × Priority Fee = 110,000 × 4 = 440,000 Gwei = **0.00044 ETH**

(b) [3 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 36 + 4 = **40 Gwei**
- **Tx2 Fee** = Gas Used × Effective Gas Price = 55,000 × 40 = 2,200,000 Gwei = **0.0022 ETH**
- **Burned** = Gas Used × Base Fee = 55,000 × 36 = 1,980,000 Gwei = **0.00198 ETH**
- **Validator Reward** = Gas Used × Priority Fee = 55,000 × 4 = 220,000 Gwei = **0.00022 ETH**

(c) [2 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 30 + 3 = **33 Gwei**
- **Tx3 Fee** = Gas Used × Effective Gas Price = 21,000 × 33 = 693,000 Gwei = **0.000693 ETH**
- **Total deducted from A** = Value + Fee = 2 + 0.000693 = **2.000693 ETH**

(d) [2 marks]

- **Total burned** = 3,520,000 + 1,980,000 + 630,000 = 6,130,000 Gwei = **0.00613 ETH**
- **Total validator reward** = 440,000 + 220,000 + 63,000 = 723,000 Gwei = **0.000723 ETH**
- Check: total fees = 6,853,000 Gwei = 0.006853 ETH = 0.00396 + 0.0022 + 0.000693

Question 6 (Reverse-type) [10 Marks]

Here the fee is given and you must work backwards.

(a) A contract deployment (**Tx1**) uses 90,000 gas. The total fee paid was **0.00378 ETH** and the Priority Fee was 2 Gwei. Find (i) the Effective Gas Price, (ii) the Base Fee, (iii) the ETH burned, and (iv) the validator reward. **[4]**

(b) The contract is updated (**Tx2**) when the Base Fee has risen to 50 Gwei. Gas Used = 48,000 and Priority Fee = 2 Gwei. Calculate the Tx2 cost in ETH. **[3]**

(c) A then sends **2 ETH** to B (**Tx3**, standard transfer). The total fee paid was **0.000483 ETH** and the Priority Fee was 3 Gwei. Find the Base Fee and the total ETH deducted from A. **[3]**

Answer

(a) [4 marks]

- Fee in Gwei = $0.00378 \times 10^9 = 3{,}780{,}000$ Gwei
- (i) **Effective Gas Price** = Fee ÷ Gas Used = 3,780,000 ÷ 90,000 = **42 Gwei**
- (ii) **Base Fee** = Effective Gas Price – Priority Fee = 42 – 2 = **40 Gwei**
- (iii) **Burned** = 90,000 × 40 = 3,600,000 Gwei = **0.0036 ETH**
- (iv) **Validator reward** = 90,000 × 2 = 180,000 Gwei = **0.00018 ETH**

(b) [3 marks]

- **Effective Gas Price** = Base Fee + Priority Fee = 50 + 2 = **52 Gwei**
- **Tx2 Fee** = Gas Used × Effective Gas Price = 48,000 × 52 = 2,496,000 Gwei = **0.002496 ETH**
- **Burned** = Gas Used × Base Fee = 48,000 × 50 = 2,400,000 Gwei = **0.0024 ETH**
- **Validator Reward** = Gas Used × Priority Fee = 48,000 × 2 = 96,000 Gwei = **0.000096 ETH**

(c) [3 marks]

- Fee in Gwei = $0.000483 \times 10^9 = 483{,}000$ Gwei
- **Effective Gas Price** = 483,000 ÷ 21,000 = 23 Gwei
- **Base Fee** = 23 – 3 = **20 Gwei**
- **Total deducted from A** = 2 + 0.000483 = **2.000483 ETH** (B receives 2 ETH)